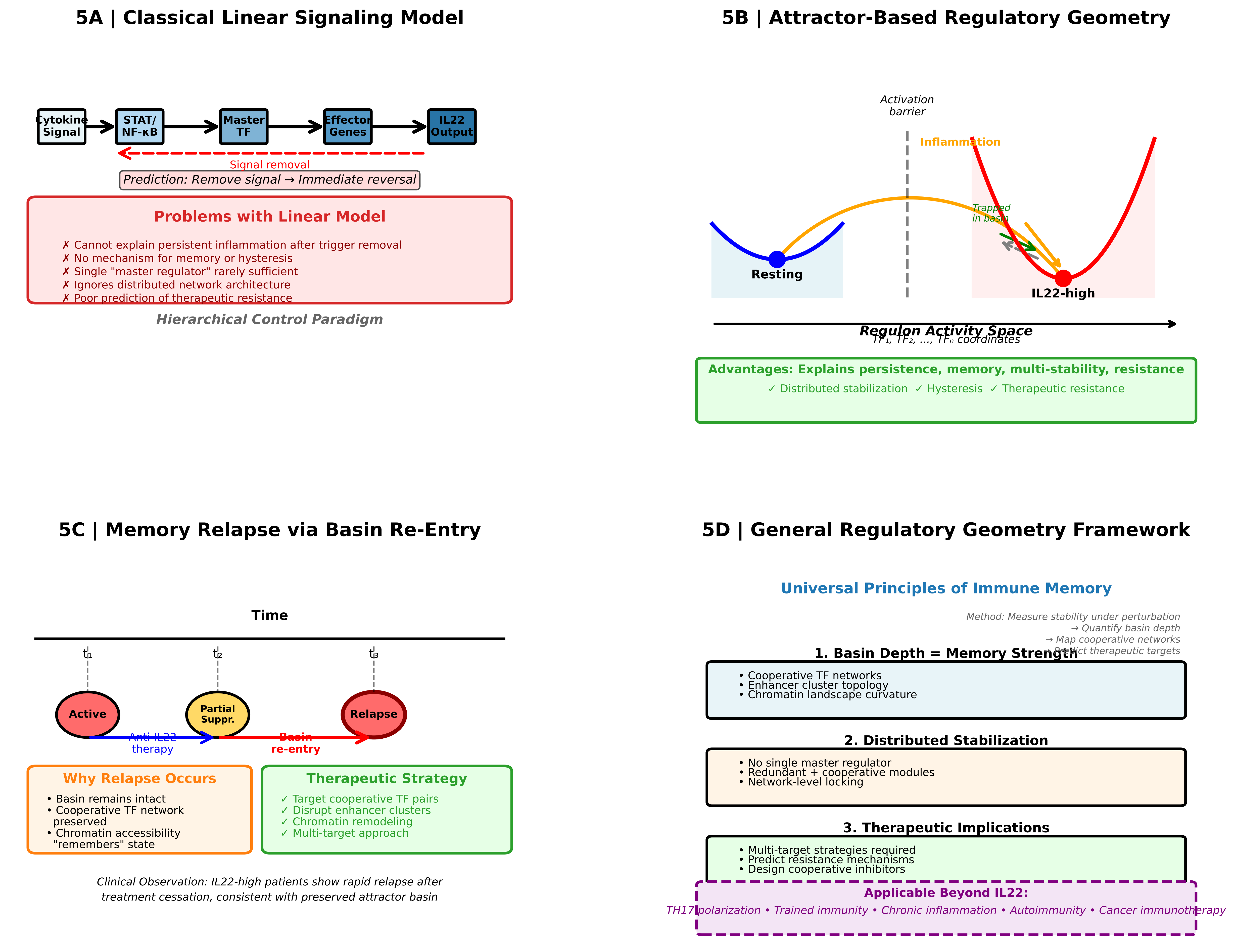
5A | Classical Linear Signaling Model
Cytokine Signal
STAT/ NF-κB
Master TF
Effector Genes
IL22 Output
Signal removal
Prediction: Remove signal → Immediate reversal
Problems with Linear Model
✗ Cannot explain persistent inflammation after trigger removal
✗ No mechanism for memory or hysteresis
✗ Single "master regulator" rarely sufficient
✗ Ignores distributed network architecture
✗ Poor prediction of therapeutic resistance
Hierarchical Control Paradigm
5B | Attractor-Based Regulatory Geometry
Activation barrier
Inflammation
Trapped in basin
Resting
IL22-high
Regulon Activity Space
TF₁, TF₂, ..., TFₙ coordinates
Advantages: Explains persistence, memory, multi-stability, resistance
✓ Distributed stabilization ✓ Hysteresis ✓ Therapeutic resistance
5C | Memory Relapse via Basin Re-Entry
Time
t₁
t₂
t₃
Active
Partial Suppr.
Relapse
Anti-IL22 therapy
Basin re-entry
Why Relapse Occurs
• Basin remains intact
• Cooperative TF network preserved
• Chromatin accessibility "remembers" state
Therapeutic Strategy
✓ Target cooperative TF pairs
✓ Disrupt enhancer clusters
✓ Chromatin remodeling
✓ Multi-target approach
Clinical Observation: IL22-high patients show rapid relapse after treatment cessation, consistent with preserved attractor basin
5D | General Regulatory Geometry Framework
Universal Principles of Immune Memory
Method: Measure stability under perturbation
→ Quantify basin depth
→ Map cooperative networks
→ Predict therapeutic targets
1. Basin Depth = Memory Strength
• Cooperative TF networks
• Enhancer cluster topology
• Chromatin landscape curvature
2. Distributed Stabilization
• No single master regulator
• Redundant + cooperative modules
• Network-level locking
3. Therapeutic Implications
• Multi-target strategies required
• Predict resistance mechanisms
• Design cooperative inhibitors
Applicable Beyond IL22:
TH17 polarization • Trained immunity • Chronic inflammation • Autoimmunity • Cancer immunotherapy